Management of TPA-Induced Angioedema

Overview

Angioedema is a rare but potentially life-threatening complication of **alteplase (tPA)** administration, occurring in up to **1–5% of patients**, especially in those with **middle cerebral artery (MCA) infarcts** or **ACE inhibitor use**.

Recognition

Time of Onset:

- Typically occurs within **1–2 hours** after tPA administration, often during or shortly after the infusion.

Presentation:

- Unilateral or bilateral **facial swelling**
- **Tongue, lips, or airway** involvement
- May be **asymmetric**, often on the side **contralateral** to the stroke
- May or may not have urticaria or rash

Early recognition is critical to prevent **airway compromise**.

Risk Factors

- **Use of ACE inhibitors**
- **Large infarct involving the insular cortex or MCA territory**
- **Female sex**
- **History of angioedema**

Management Protocol

◇ Step 1: Stop the Infusion

- Immediately **discontinue alteplase** if angioedema is suspected or observed.

◇ Step 2: Secure the Airway

- Assess for signs of airway obstruction
- Call anesthesia or ENT early if airway involvement is suspected
- Prepare for **emergent intubation** if progressive swelling or stridor

◇ Step 3: Administer Medications

- **IV Corticosteroids:**
  - **Methylprednisolone 125 mg IV** once
- **IV Antihistamines:**
  - **Diphenhydramine 50 mg IV**
  - **Famotidine 20 mg IV** (H2 blocker)
- **Epinephrine (if systemic symptoms or airway involvement):**
  - **0.3–0.5 mg IM** (1:1000) every 5–15 minutes as needed
  - Or **0.1 mg IV push** (1:10,000) for hypotension or airway compromise

Step 4: Supportive Care & Monitoring

- Close monitoring in the **ICU or ED resuscitation area**
- Frequent reassessment of swelling and respiratory status
- Consider **C1 esterase inhibitor (Berinert)** or **Icatibant (Firazyr)** for refractory cases (based on institutional availability and allergist consultation)

Documentation & Reporting

- Document time of tPA administration, time of onset, symptoms, and treatments given
- Report to institutional safety reporting systems and stroke registry, if applicable

Pathophysiology (Optional Education Add-on)

- Believed to be **bradykinin-mediated**, not histamine-driven
- Explains why **ACE inhibitors**, which increase bradykinin, raise the risk

References

1. Hill MD, Lye T, Moss H, et al. Hemi-orolingual angioedema and ACE inhibition after alteplase treatment of stroke. *Neurology*. 2003;60(9):1525–1527. doi:10.1212/01.WNL.0000063315.36590.EE
2. Suri MF, Nasar A, Qureshi AI. Prevalence and outcomes of orolingual angioedema in acute ischemic stroke patients treated with thrombolysis. *Stroke*. 2006;37(12):e155–e157. doi:10.1161/01.STR.0000252133.15476.54
3. Powers WJ, Rabinstein AA, Ackerson T, et al. 2019 Guidelines for the Early Management of Acute Ischemic Stroke. *Stroke*. 2019;50(12):e344–e418. doi:10.1161/STR.0000000000000211